

LAB 5: CHROMATOGRAPHIC SEPARATION OF PLANT PIGMENTS

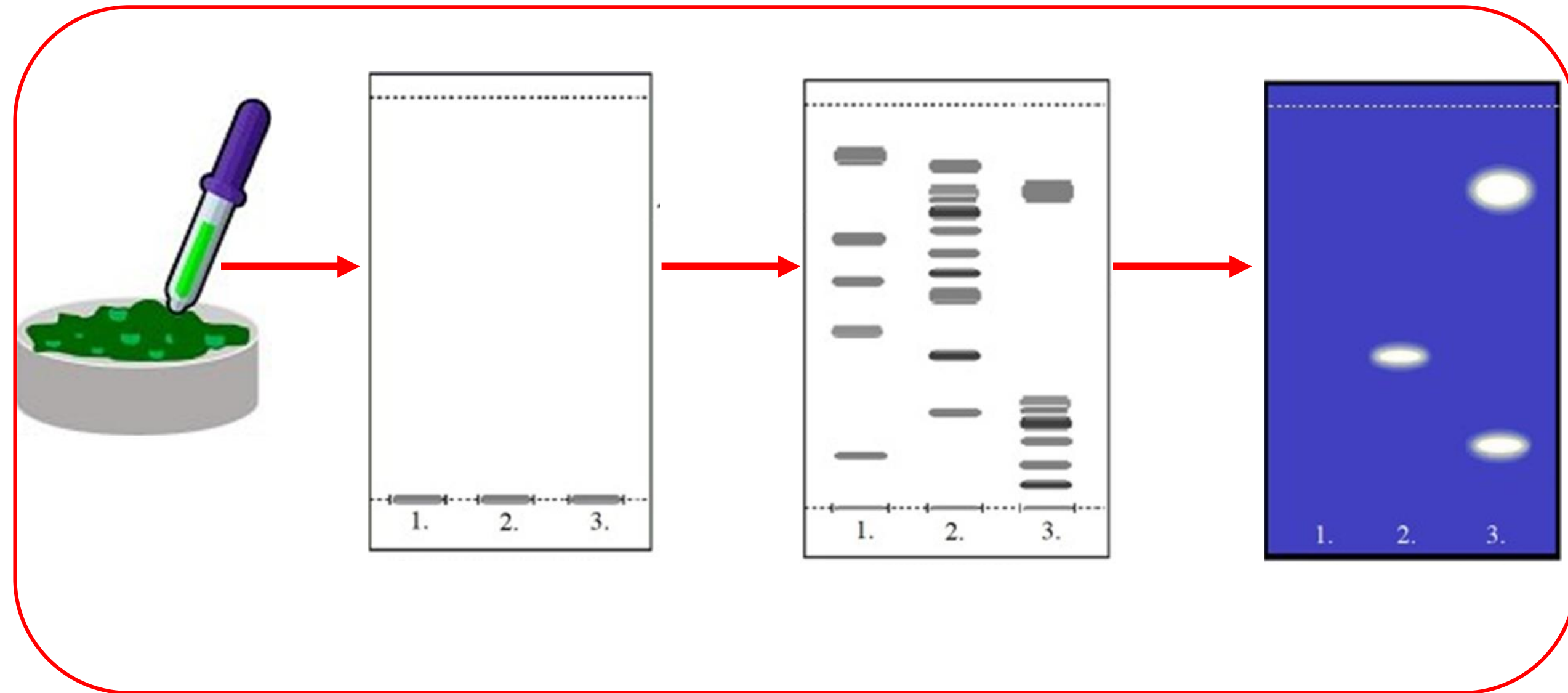
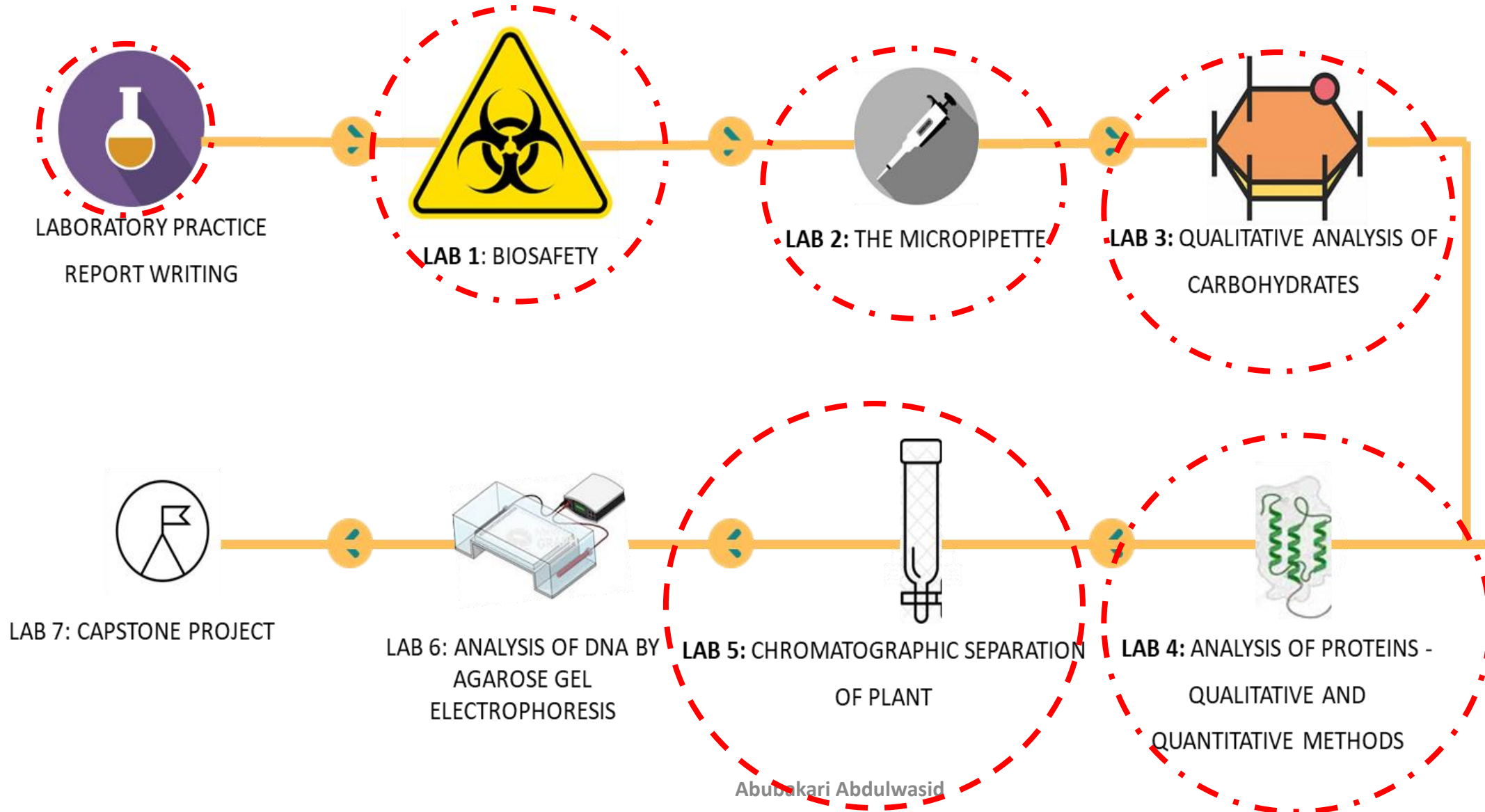
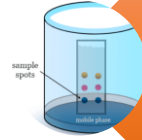


Table of content for the Course



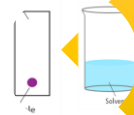
This week Lecture



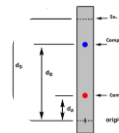
Define the fundamental principles of chromatography.



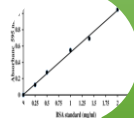
Explain the specific mechanism of Thin Layer Chromatography (TLC)



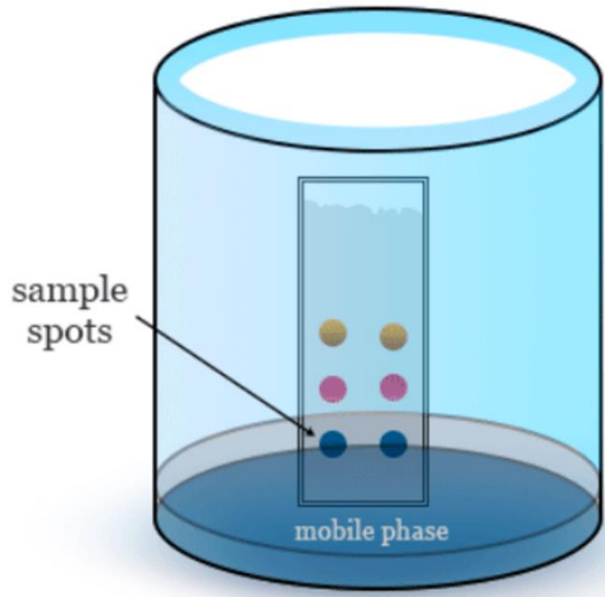
Identify the stationary and mobile phases for separating plant pigments



Calculate and interpret the Retention Factor (R_f) value



Outline the procedure for performing a TLC experiment & analyze a chromatogram



What is Chromatography?

The Fundamental Principle: Separation

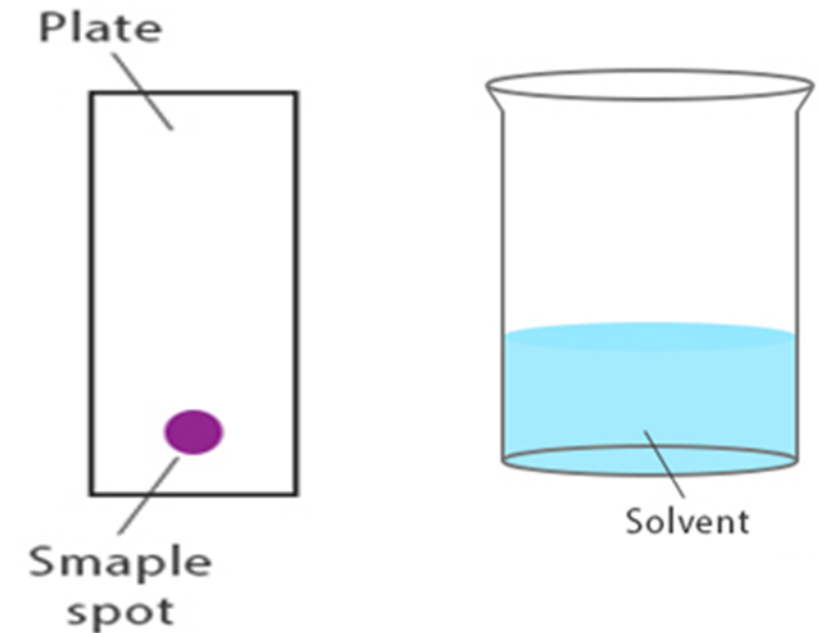
Chromatography is a technique used to separate the components of a mixture.

This separation is achieved by distributing the components between two phases:

❖ **Stationary Phase:** A fixed material (solid or liquid) that stays in place.

❖ **Mobile Phase:** A fluid (liquid or gas) that moves through the stationary phase.

The core principle is differential partitioning: Components that interact more strongly with the **stationary phase** move slowly, while components that are more soluble in the **mobile phase** move faster.



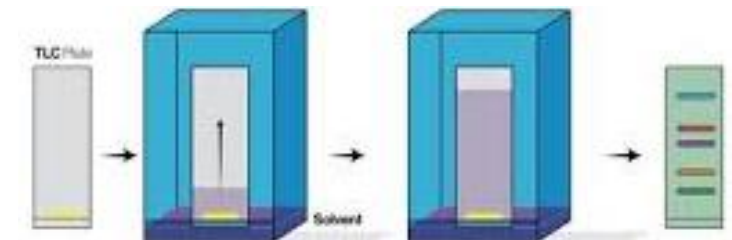
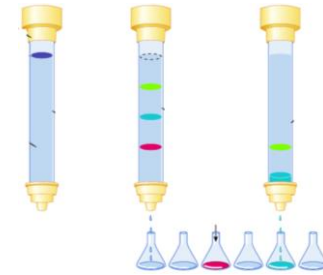
Types of Chromatography



A Versatile Family of Techniques

While we focus on TLC, chromatography is a broad field:

- ❖ **Column Chromatography**: Stationary phase is packed in a vertical column.
- ❖ **Gas Chromatography (GC)**: Mobile phase is a gas; used for volatile compounds.
- ❖ **High-Performance Liquid Chromatography (HPLC)**: A high-pressure, high-resolution version of column chromatography.
- ❖ **Thin-Layer Chromatography (TLC)**: Stationary phase is a thin layer on a flat plate. **(Our Focus)**.



Thin Layer Chromatography (TLC)

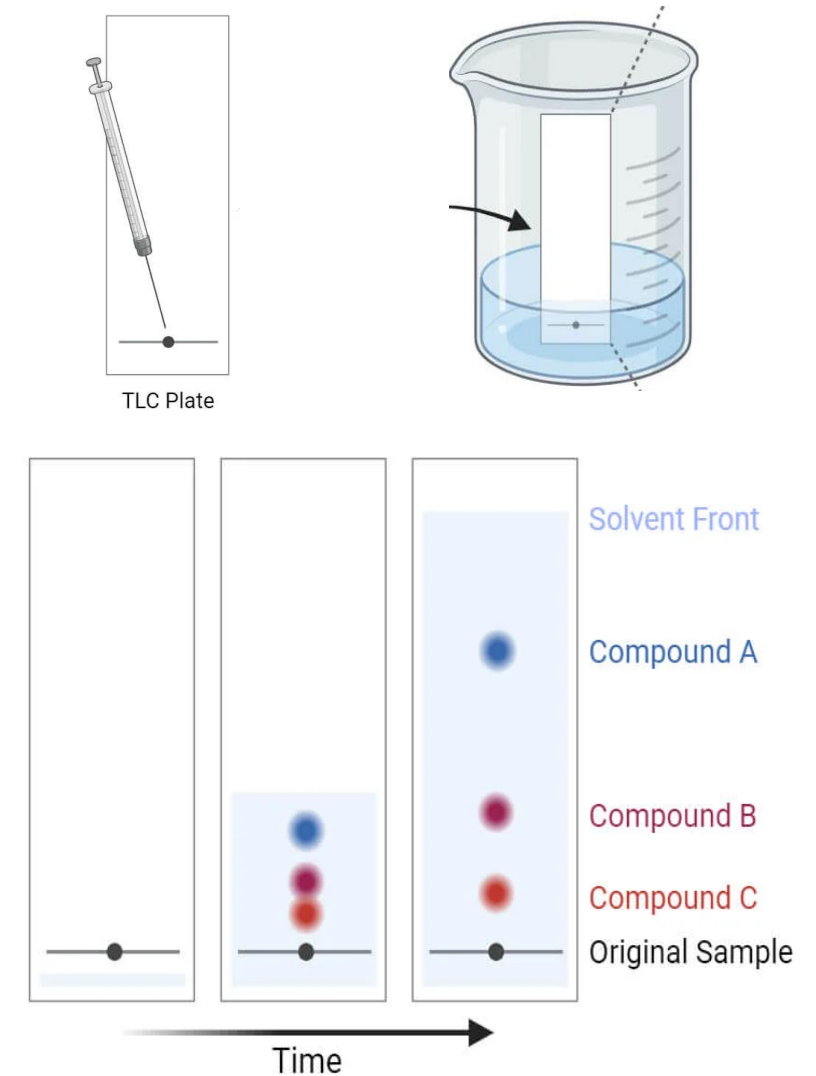
TLC is a type of adsorption chromatography:

❖ **Stationary Phase:** A thin layer of an adsorbent material, like silica gel, coated onto a plate.

Silica gel is highly polar.

❖ **Mobile Phase:** A solvent or mixture of solvents (e.g., Petroleum ether: Acetone). In our lab, this solvent is largely non-polar.

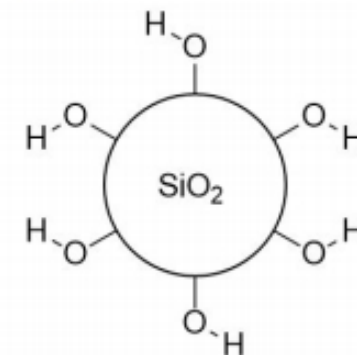
It is excellent for quick qualitative analysis, such as identifying compounds or checking reaction progress.



The Stationary Phase: Silica Gel

The "Sticky" Surface:

- ❖ Our stationary phase is a thin plate coated with silica gel.
- ❖ The surface of silica gel is covered with silanol groups (-OH).
- ❖ These -OH groups make the surface extremely polar and capable of forming hydrogen bonds.

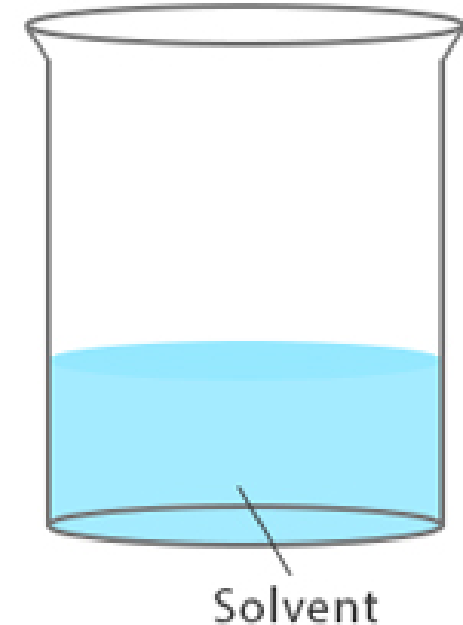


This is what "polar" molecules in our mixture will stick to.

The Mobile Phase: The Driving Force

The mobile phase is the solvent system that climbs the plate via capillary action, carrying the sample with it.

- ❖ **Its function:** To dissolve the sample and compete with the stationary phase for the molecules.
- ❖ **The Rule:** For our polar silica plate, we **use a non-polar mobile phase**. **Non-polar compounds** in the sample prefer to dissolve in the **mobile phase** and therefore travel much further up the plate (High R_f).

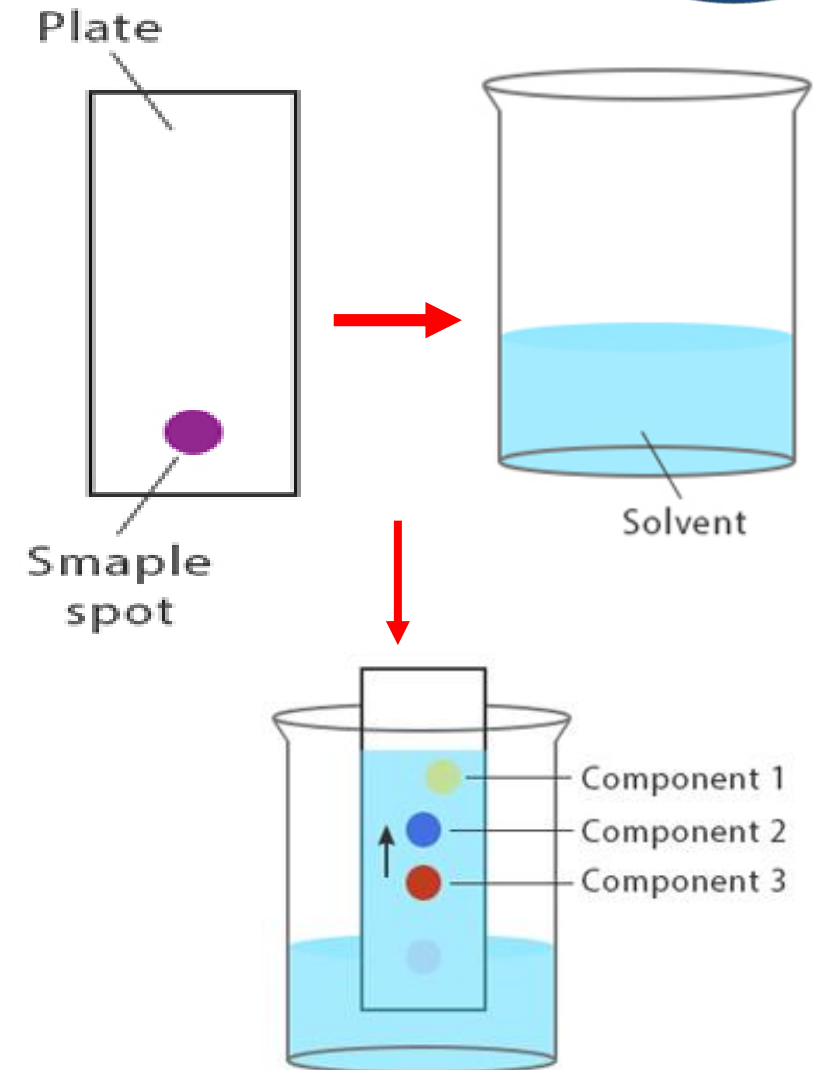


Hexane (Non-polar) -> Toluene -> Dichloromethane -> Acetone -> Ethanol -> Water (Polar)

The Principle of Separation in TLC

A Competition Based on Polarity:

- ❖ The **sample** is spotted on the **origin**.
- ❖ The **non-polar mobile phase** begins to **move up** the plate.
- ❖ **Polar molecules** in the sample are **strongly attracted** to the **polar silica gel (stationary phase)**. They "stick" and move very little.
- ❖ **Non-polar molecules** are not attracted to the silica gel. They **dissolve readily in the mobile phase** and are carried much further up the plate.



Quantifying Separation: The Retention Factor (Rf)

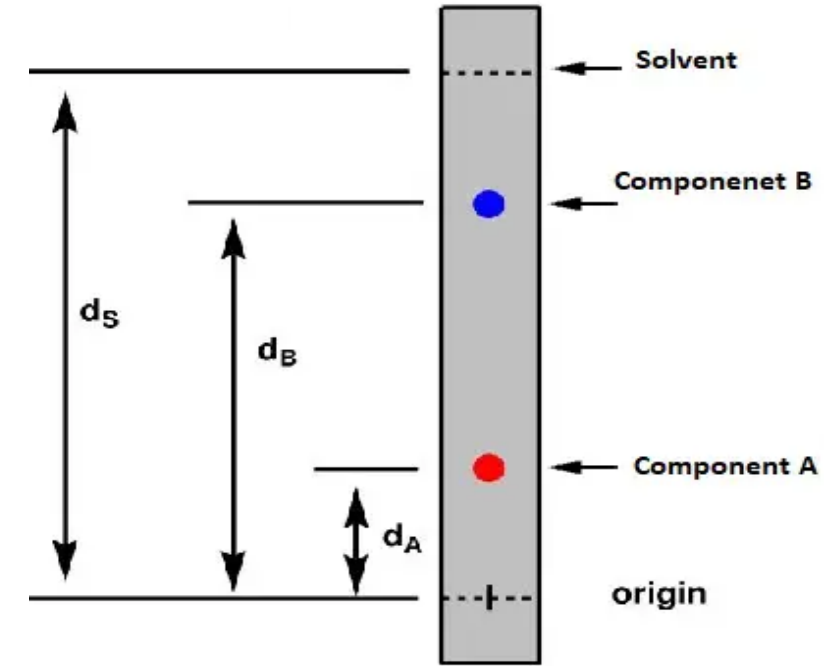


Quantifying the Separation

The **Retention Factor (Rf)** is a ratio that quantifies the movement of a separated component.

$$R_f = \frac{\text{Distance travelled by the component}}{\text{Distance travelled by the solvent front}}$$

- ❖ The Rf value is always between **0 and 1**.
- ❖ **Low Rf Value (e.g., 0.2)**: Indicates a polar compound.
- ❖ **High Rf Value (e.g., 0.8)**: Indicates a non-polar compound.



R_f of component A =

$$\frac{d_A}{d_S}$$

R_f of component B =

$$\frac{d_B}{d_S}$$

Case Study: Separating Plant Pigments



Applying the Principle:

We will separate pigments from plant leaves using a non-polar mobile phase

❖ The Pigments (from most to least polar):

- ✓ Chlorophyll b: Most polar
- ✓ Chlorophyll a
- ✓ Xanthophylls
- ✓ Carotenes: Least polar (it's a hydrocarbon!)



Thinking Like a Scientist: Predicting a Chromatogram

Based on polarity, predict the final chromatogram:

Which pigment will have the highest R_f value (travel the furthest)?

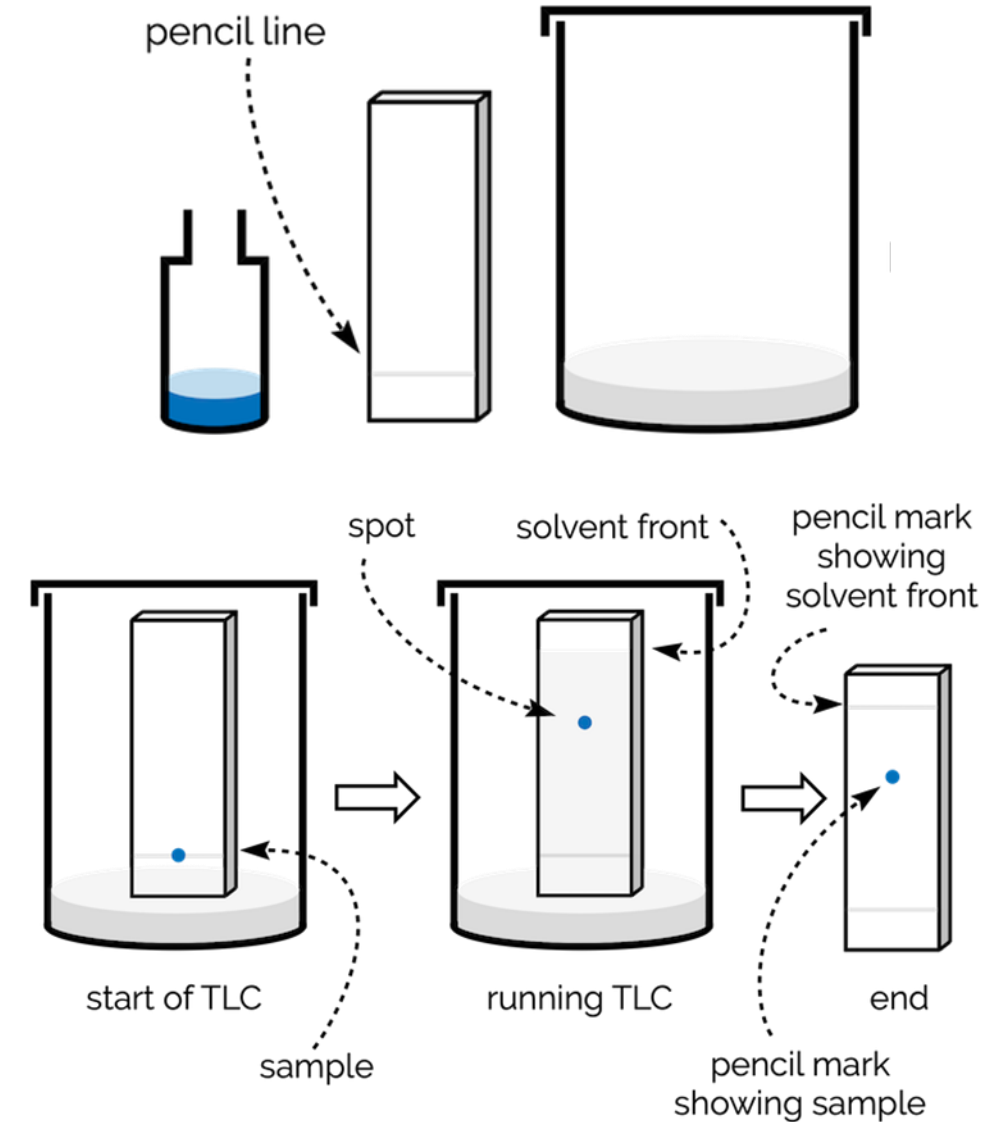
✓ **Answer: Carotene**, because it is the most non-polar.

Which pigment will have the lowest R_f value (travel the least)?

✓ **Answer: Chlorophyll b**, because it is the most polar.

Procedure for TLC

- ❖ **Prepare the Plate:** Use a PENCIL to lightly draw an origin line.
- ❖ **Spot the Sample:** Use a capillary tube to apply a small, concentrated spot. Let it dry.
- ❖ **Prepare the Chamber:** Add mobile phase (level must be below the origin line). Cover and saturate.
- ❖ **Develop the Plate:** Place the plate in the chamber. Let the solvent front move up.
- ❖ **Visualize and Mark:** Immediately remove and mark the solvent front and spots with a pencil.



Thinking Like a Scientist: Troubleshooting

❖ **Problem:** Spots are streaky, not round.

✓ **Cause:** The initial spot was too large or too concentrated.



❖ **Problem:** All spots remained at the origin.

✓ **Cause:** The mobile phase is too non-polar. It doesn't have the "power" to move even the least polar component.



❖ **Problem:** All spots moved to the solvent front.

✓ **Cause:** The mobile phase is too polar. It moves all components, regardless of their polarity.

Applications of Chromatography



Why is this technique so important?

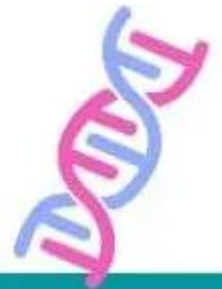
- ❖ **Pharmaceutical Industry:** To determine the purity of a drug sample or to isolate active compounds.
- ❖ **Forensics:** To analyze ink samples or identify unknown substances from a crime scene.
- ❖ **Food Industry:** To separate and identify additives, preservatives, or contaminants in food products.
- ❖ **Biochemical Research:** To monitor the progress of a chemical reaction by observing the disappearance of reactants and the appearance of products.



Pharmaceuticals



Forensic
Science



Life-Sciences

TLC Safety & Waste Disposal



❖ **Chemical Safety:** Solvents (Acetone, Petroleum Ether) are flammable and have potent fumes. Work in a well-ventilated area or a fume hood.

❖ **Handling:** Handle TLC plates by the edges only.

❖ **Waste Disposal:** Dispose of used solvents in the "Organic/Solvent Waste" container.



Key Takeaways

- ❖ Chromatography separates molecules based on partitioning between a stationary and mobile phase.
- ❖ TLC separates based on polarity.
- ❖ Polar compounds stick to the polar silica gel (Low R_f).
- ❖ Non-polar compounds travel with the non-polar solvent (High R_f).

Practice Question

You are running two separate experiments on a plant extract.

a) You perform a **Thin Layer Chromatography (TLC)** separation using **a polar silica gel plate**. After development, you find two pigments: **Pigment A with an R_f value of 0.75** and **Pigment B with an R_f value of 0.30**. Which pigment (A or B) is more polar? Justify your answer based on its interaction with the stationary and mobile phases.



Questions?